

HW1- Computer Simulation II

Schewtschik, Guilherme (3748052)

April 19, 2026

1

In the first attempt, the Metropolis algorithm was used, but it became clear that for a lattice larger than 8×8 the run time around the critical temperature became too large, so instead the Wolff algorithm was chosen. With this we arrive at the following graph:

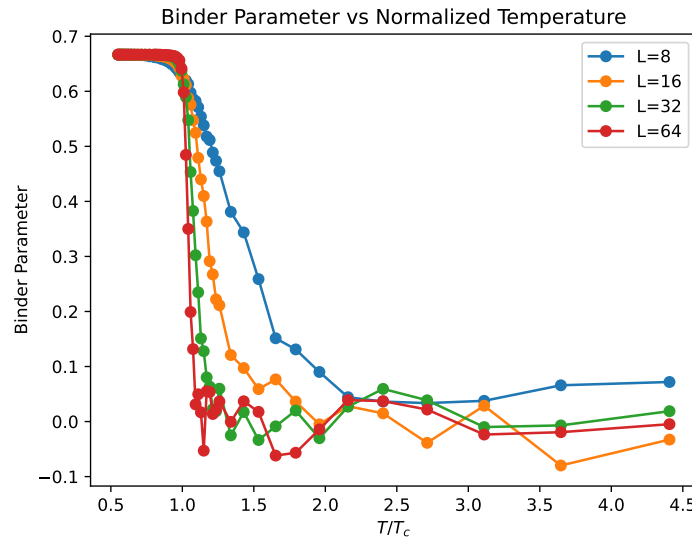


Figure 1: BinderXT

We can see that the value for U stays the similar for all L until reaching the critical temperature where they diverge.

Plotting the difference between the data points we get:

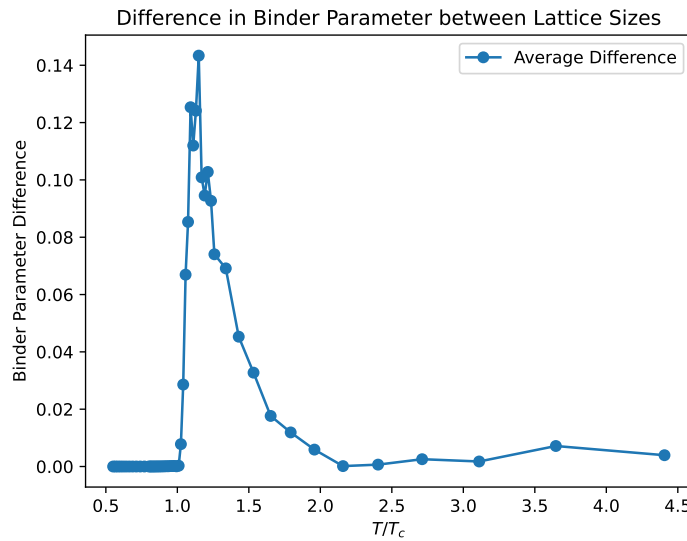


Figure 2: Caption

We can also see to a lower extent a convergence at higher temperatures.

The estimated value for U at T/T_c is:

Estimated U at T_c : $L=8$: 0.630, $L=16$: 0.628, $L=32$: 0.637, $L=64$: 0.641
Mean U at T_c : 0.634

2

Calculating the value of χ we can clearly see the peaks around the critical temperature:

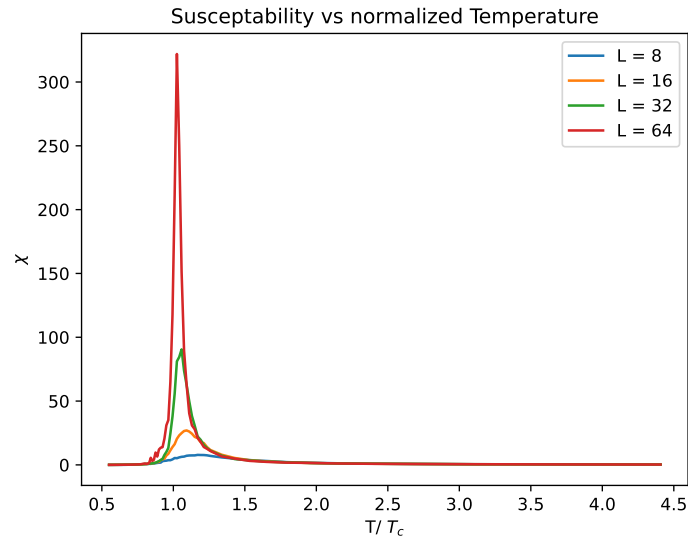


Figure 3: χ values

Taking the maximum of χ and plotting it against the lattice size L , we can see the relation between them appears to be exponential:

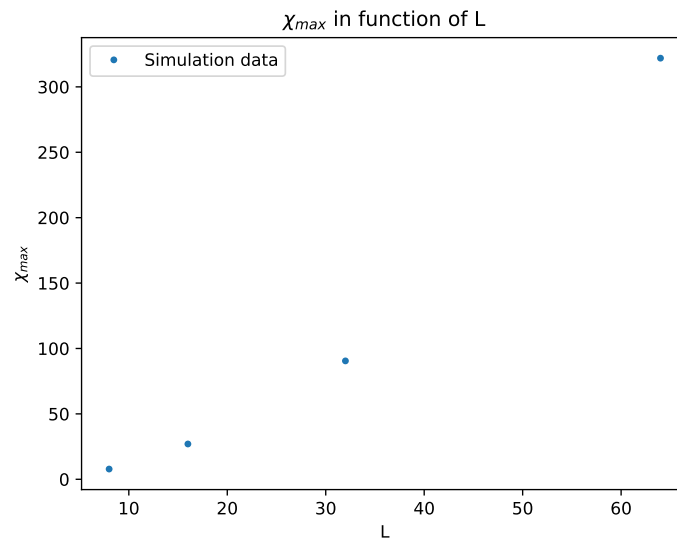


Figure 4: χ in function of L

For us to test the finite-size scaling Ansatz, we first need to make some adjustments, instead of using the given formula we shall work in log space:

$$\ln(\chi) = \frac{\gamma}{\nu} \ln(L) + c \quad (1)$$

Where c is a proportionality constant. Using this equation and fitting a line we can find the values for γ/ν and c .

$$\frac{\gamma}{\nu} = 1.783169... \quad c = -1.65374 \quad (2)$$

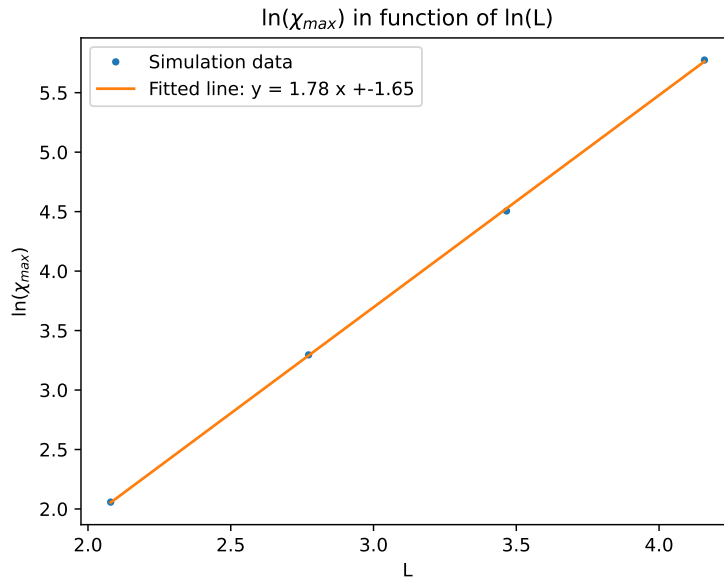


Figure 5: Fitted χ_{max}

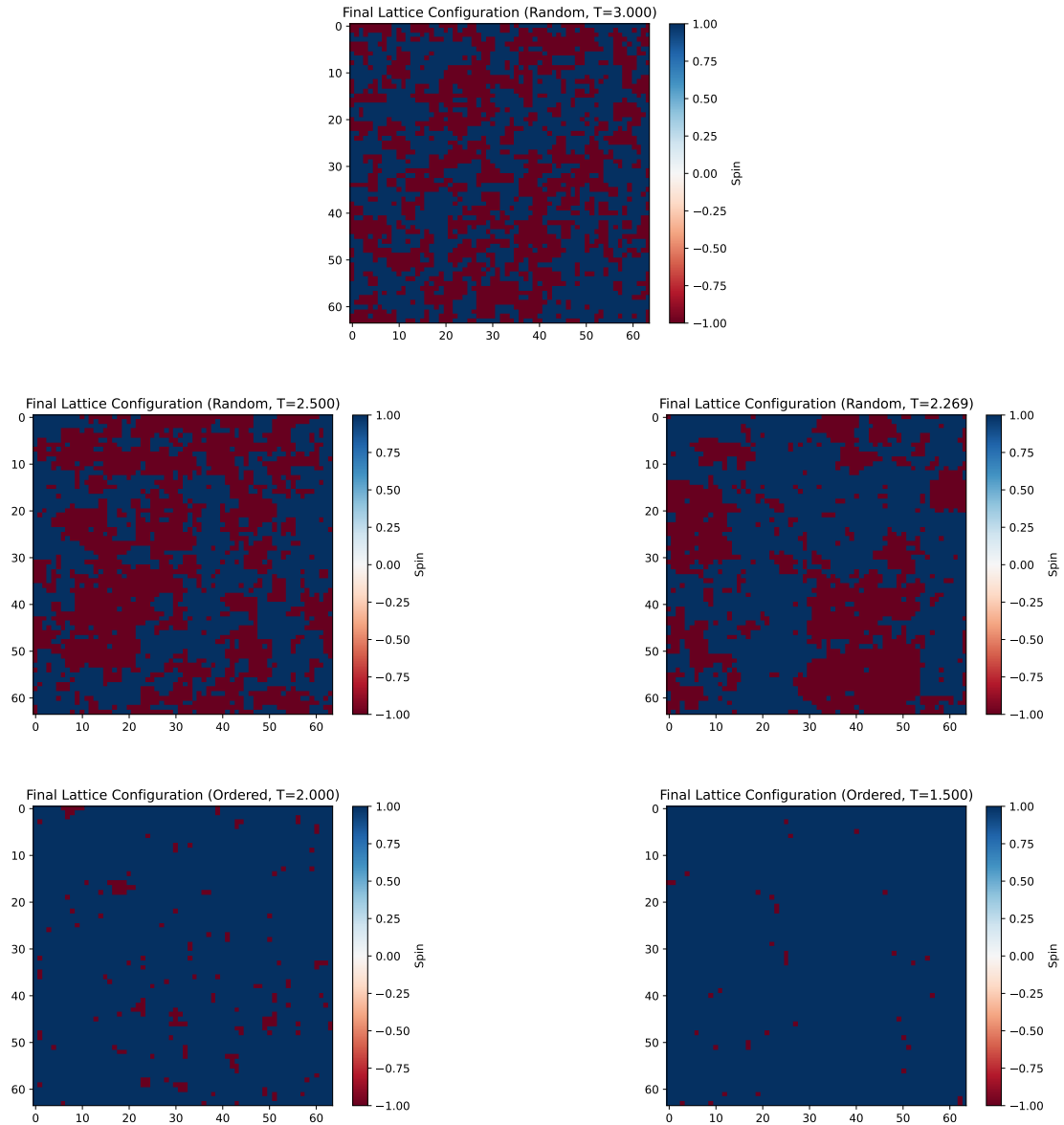
HW2- Computer Simulation II

Schewtschik, Guilherme (3748052)

April 28, 2026

1

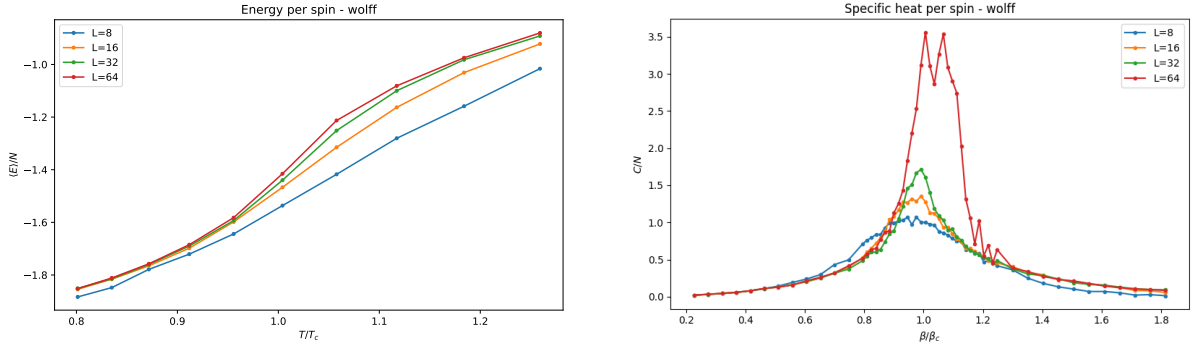
For this task the spins are plotted in a 2d grid, where the color of the cell represents the value of the spin.



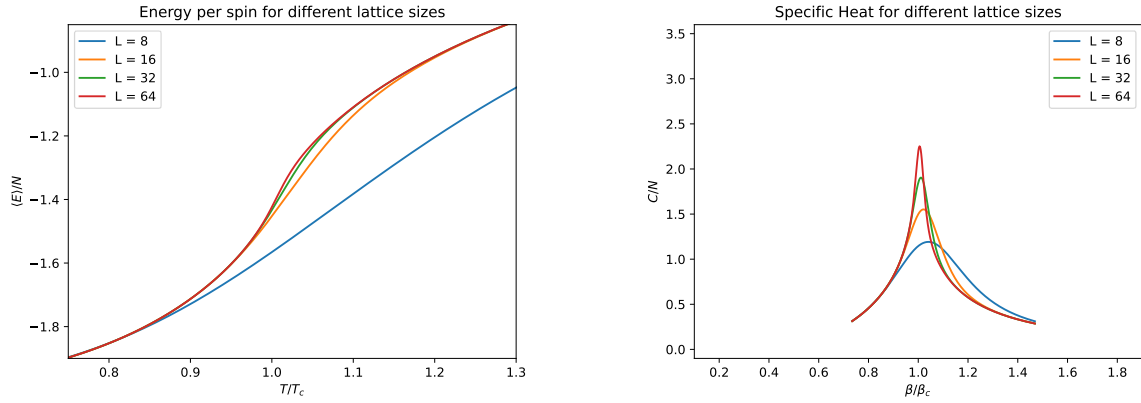
We can see the alignment of the spin directions with the lowering of temperature which is the expected result. The first three plots show a global net zero magnetization with only local alignment, while the last two show a global net magnetization with all spins aligned in the same direction.

2

Utilizing the same Wolff algorithm implemented in the previous homework, we arrive at the following graphs for the Energy per spin E , and the specific heat C around the critical temperature T_c .



We can also plot the values obtained from the module homepage and get:



We can clearly see that the values obtained from our simulation are in good agreement with the values obtained from the module homepage only presenting some minor deviations, specially for the heat capacity, where the peak value for the $L = 64$ grid surpasses the expected result by quite a lot. We can also see that the specific heat has a peak around the critical temperature T_c , which is a characteristic feature of phase transitions.

We estimate the scaling of the autocorrelation time near the critical temperature using the ansatz

$$\tau(L) \sim L^z.$$

A linear fit in log-log space is used to extract the dynamical exponent z .

For the Wolff (cluster) algorithm, we obtain:

- Energy:

$$z_E = -0.70 \pm 0.56, \quad R^2 = 0.43$$

- Magnetization:

$$z_M = -0.99 \pm 1.01, \quad R^2 = 0.32$$

HW3- Computer Simulation II

Schewtschik, Guilherme (3748052)

April 26, 2026

1

Using the code from the first homework (where I already had implemented the Wolff algorithm for improvements in computational time) we can easily calculate $dU/d\beta$ at β_c :

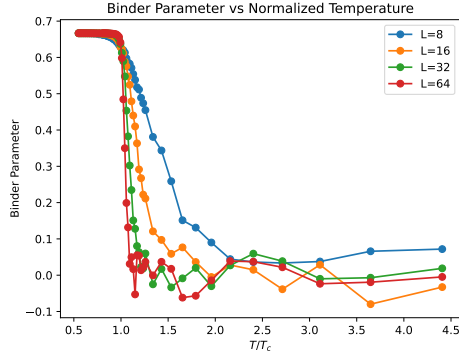


Figure 1: Binder parameter

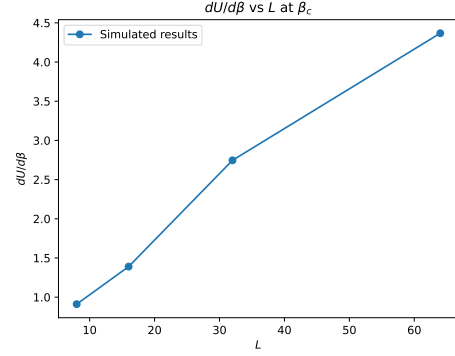
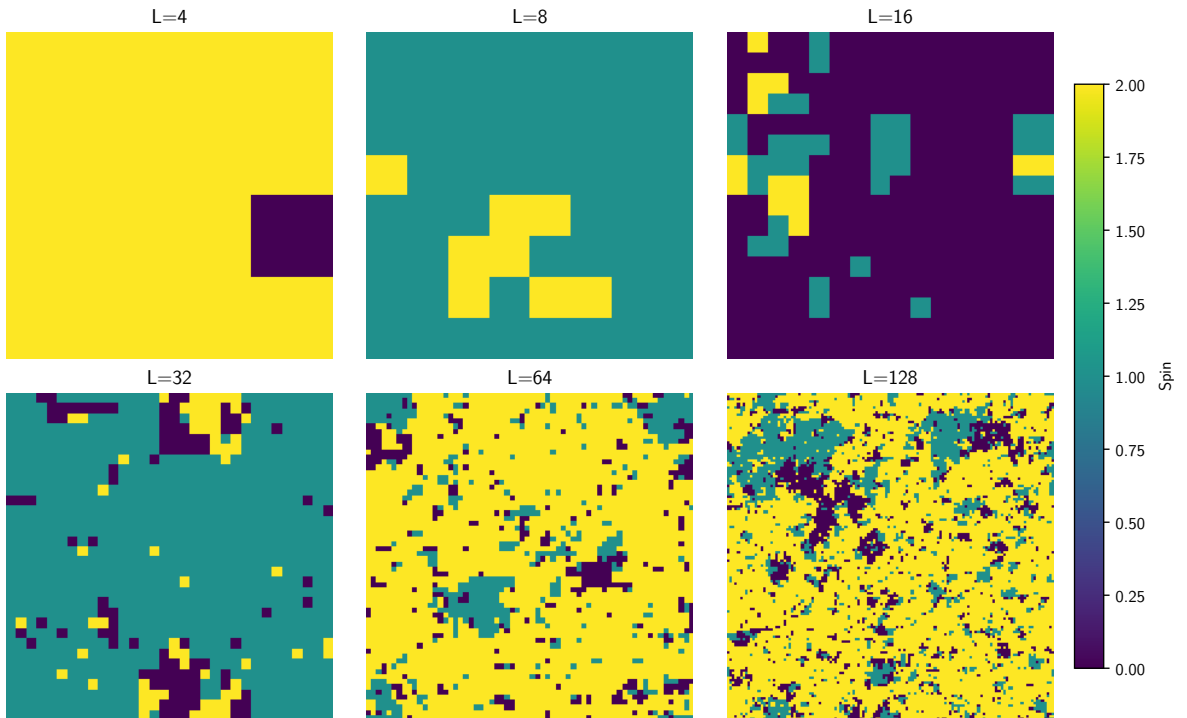


Figure 2: derivative of the Binder parameter

We can see that $dU/d\beta$ scales linearly with lattice size.

2

Using the q -states Potts model and $q = 3$, $\beta = \ln(1 + \sqrt{q})$, we can analyze the behavior of the system for different lattice sizes L . The following graph illustrate the equilibrium state of the system for different lattice sizes L :



We can see the distinct phases that arise at the critical temperature, we can further calculate the Energy and Specific heat of the lattices. Plotting it in function of lattice size L :

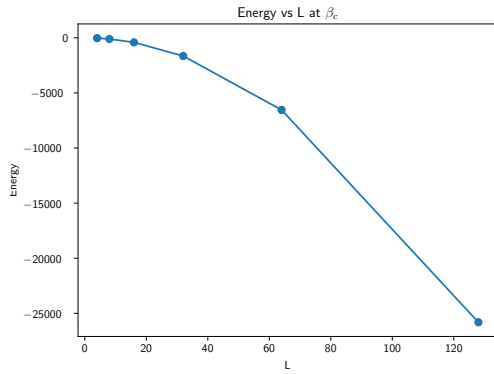


Figure 3: Energy

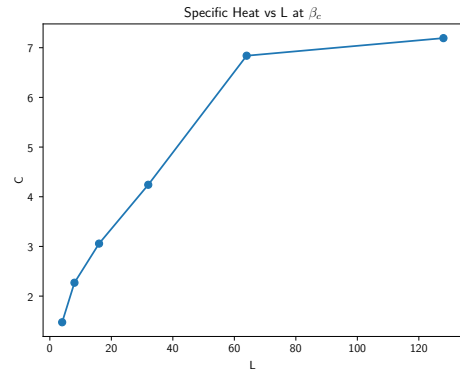


Figure 4: Specific Heat

We can see that the energy and specific heat seem to be dependent on powers of L , let's see how this graphs look in a log-scale:

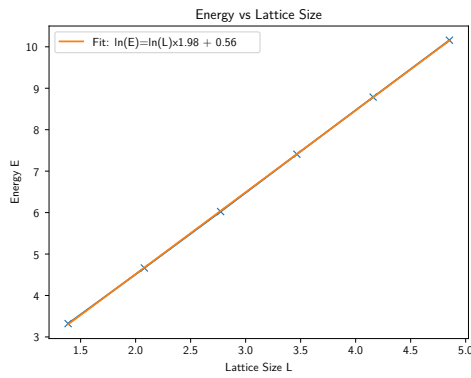


Figure 5: Absolute Energy

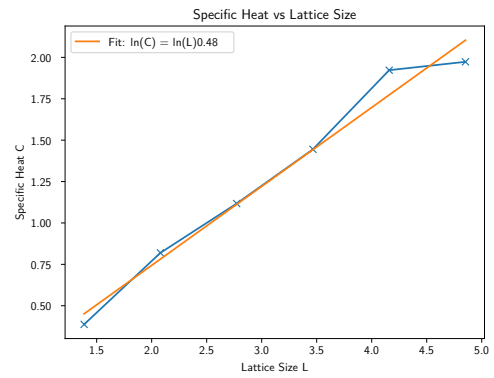


Figure 6: Specific Heat

So we see the relations:

$$E \propto -L^{1.98} \quad \text{and} \quad C \propto L^{0.48}$$

HW4- Computer Simulation II

Schewtschik, Guilherme (3748052)

June 30, 2026

1

We can calculate the autocorrelation time using the magnetization, plotting it in a log scale we get:

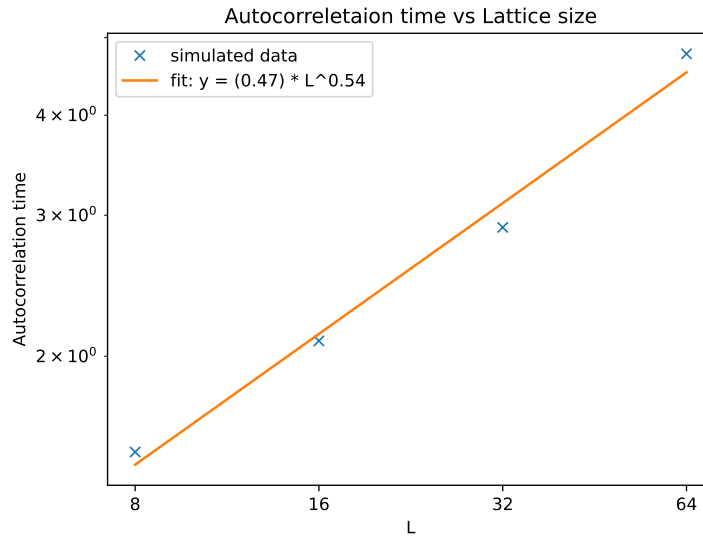


Figure 1: Autocorrelation time for the clusters

To convert this into units of lattice sweeps we use:

$$\tau_{\text{sweep}} = \tau_{\text{cluster}} \cdot \frac{\langle s \rangle}{L^2}. \quad (1)$$

Where $\langle s \rangle$ is the average cluster size. Plotting in log scale we get:

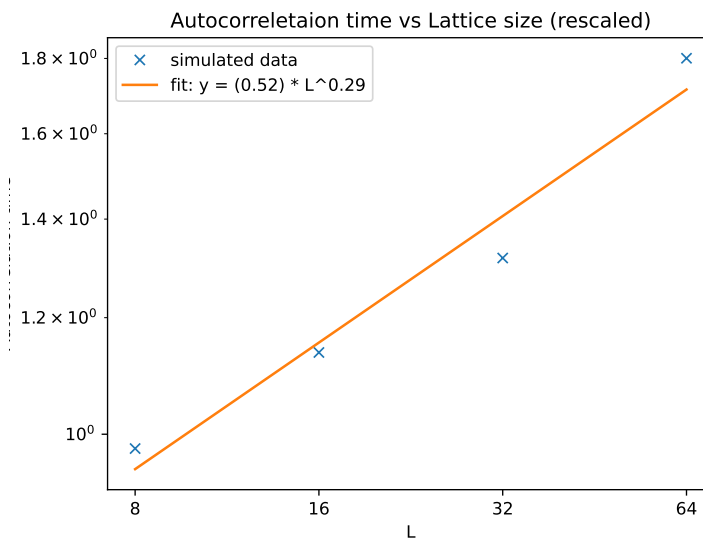


Figure 2: Rescaled correlation time

So we can see:

$$\tau \propto L^{0.29}$$

This is significantly less then the autocorrelation time of the Metropolis algorithm $\tau \propto L^{2.17}$, but still a bit larger then the expected amount for the Wolff algorithm $z \approx 0.13$.

2

Calculating the value of χ we can clearly see the peaks around the critical temperature:

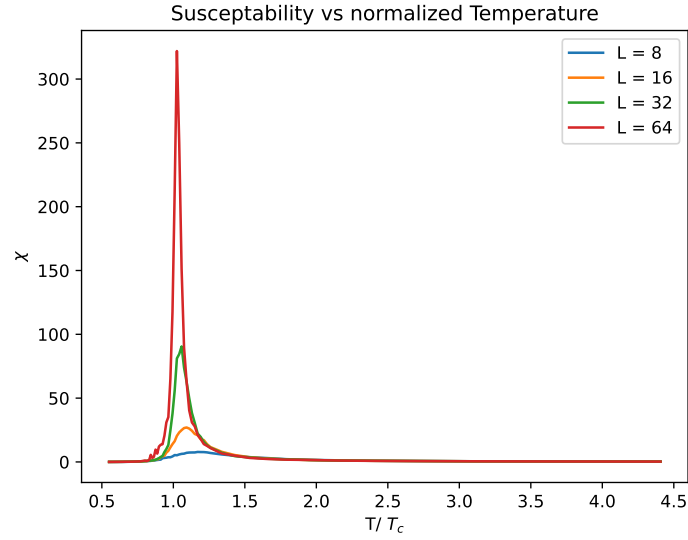


Figure 3: χ values

Taking the maximum of χ and plotting it against the lattice size L , we can see the relation between them scales by a power law:

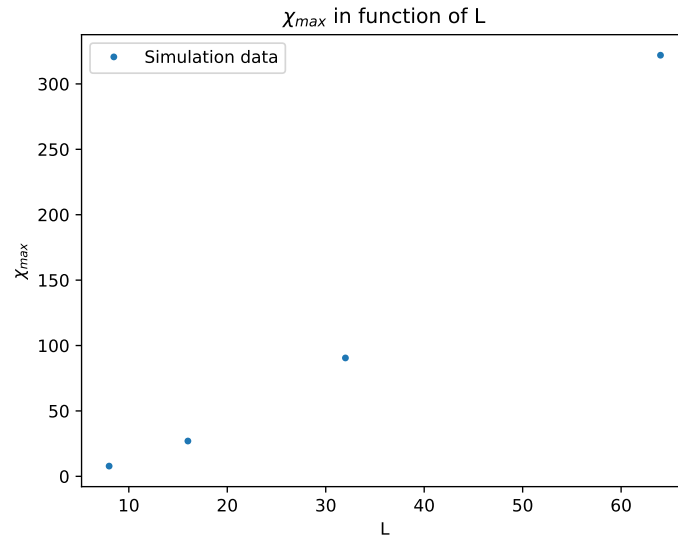


Figure 4: χ in function of L

For us to test the finite-size scaling Ansatz, we first need to make some adjustments, instead of using the given formula we shall work in log space:

$$\ln(\chi) = \frac{\gamma}{\nu} \ln(L) + c \quad (2)$$

Where c is a proportionality constant. Using this equation and fitting a line we can find the values for γ/ν and c .

$$\frac{\gamma}{\nu} = 1.783169... \quad c = -1.65374 \quad (3)$$

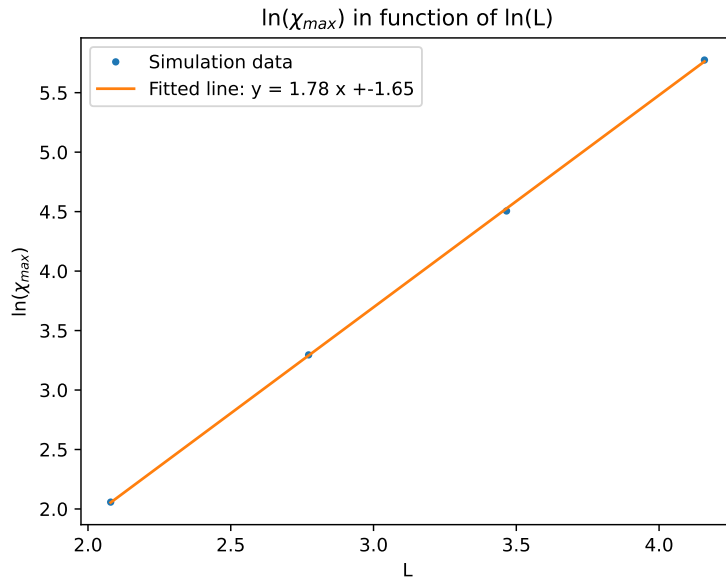


Figure 5: Fitted χ_{max}

HW6 - Computer Simulation II

Schewtschik, Guilherme (3748052)

May 26, 2026

1

Generalizing the wolff algorithm to the XY model we can measure the mean energy per site, specific heat, and susceptibility for: $\beta \in [0, 2]$ in steps of 0.05 for 8×8 and 24×24 lattices

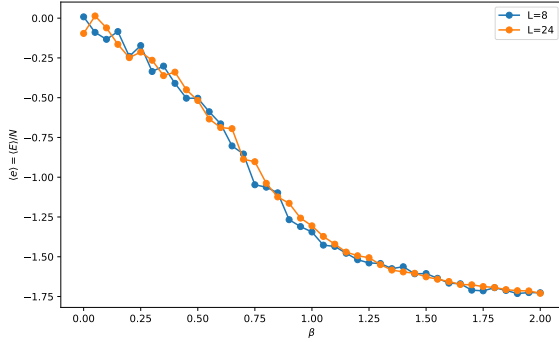


Figure 1: Mean energy per site $\times \beta$.

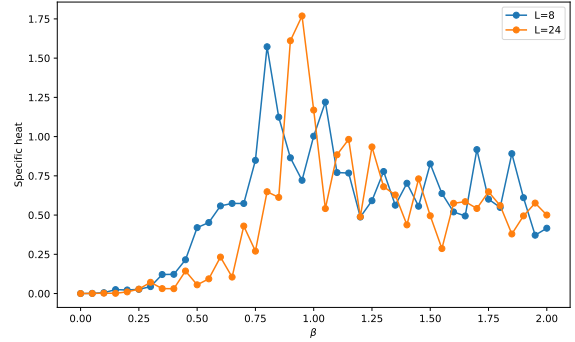


Figure 2: Specific heat $\times \beta$.

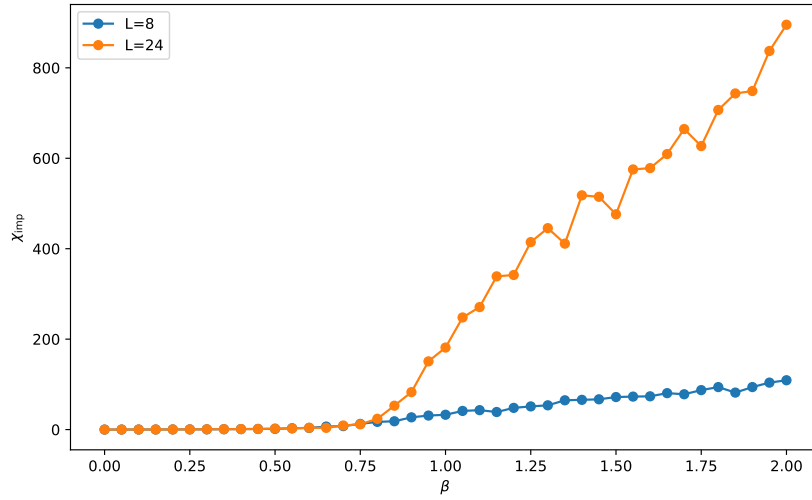


Figure 3: susceptibility $\times \beta$.

We can see the energy per site decreases with β (as expected) for both lattice sizes, the specific heat peaks around $\beta = 1$, and the susceptibility increases with β , showing a more pronounced increase around $\beta = 1$ for the larger lattice, indicating critical behavior.

2

For the 24×24 lattice, we compared the standard and improved susceptibility estimators at $\beta = 1.0, 0.9, 0.8, 0.7, 0.6$, and 0.5 .

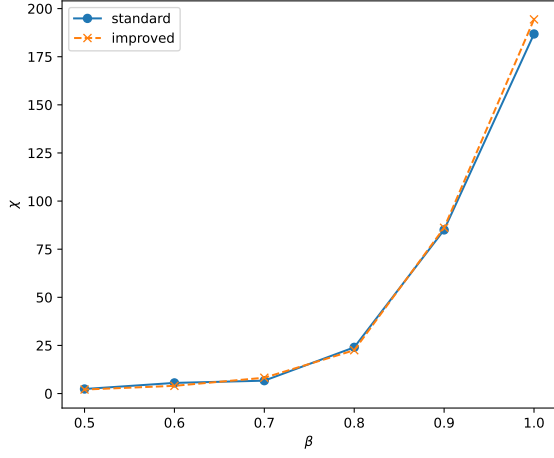


Figure 4: Standard x improved susceptibility.

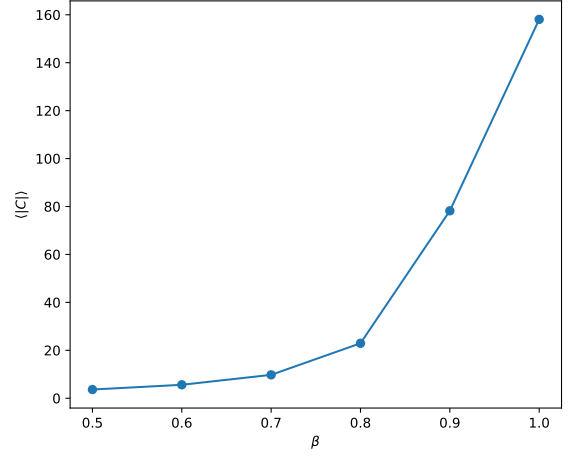


Figure 5: Mean cluster size at the comparison points.

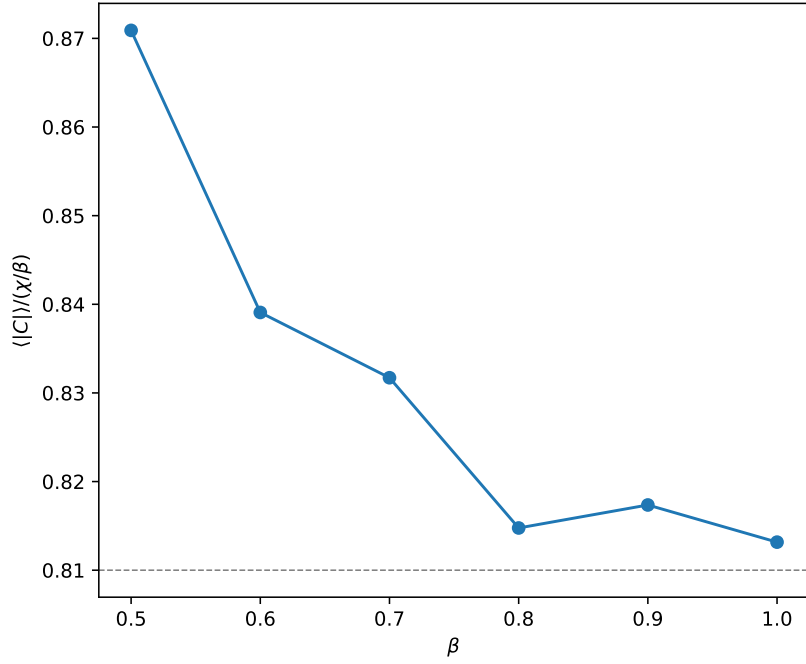


Figure 6: Ratio of mean cluster size to the improved susceptibility.

The improved susceptibility estimator remains close to the standard estimator being only slightly higher for bigger β values, the mean cluster size increases rapidly with β , and the ratio between the mean cluster size and the susceptibility remains above the expected 0.81, for small β values ranging from 0.87 to 0.82 only really approaching the expected value for smaller β , due to the bigger correlation lengths.

HW7 - Computer Simulation II

Schewtschik, Guilherme (3748052)

June 1, 2026

1 Cluster estimator for the correlation function

For the 2D Ising model we compared the standard estimator

$$G(R) = \langle s_i s_j \rangle$$

with the cluster estimator

$$\hat{G}(i, j) = \frac{V}{|C|} \Theta_C(i) \Theta_C(j),$$

using the Wolff single-cluster algorithm for lattices of size $L = 64$ and $L = 128$.

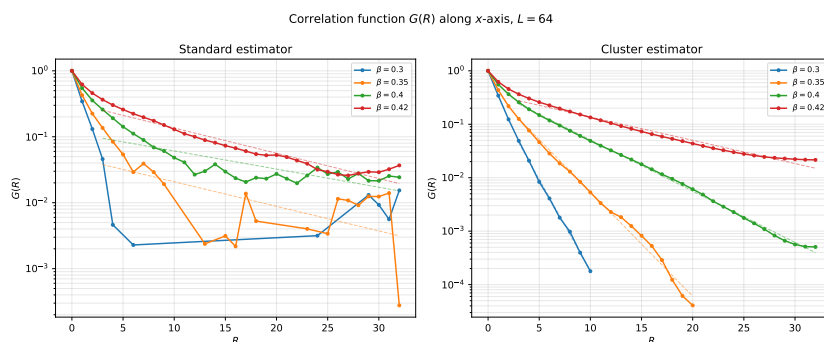


Figure 1: Correlation function $G(R)$ along the x -axis for $L = 64$.

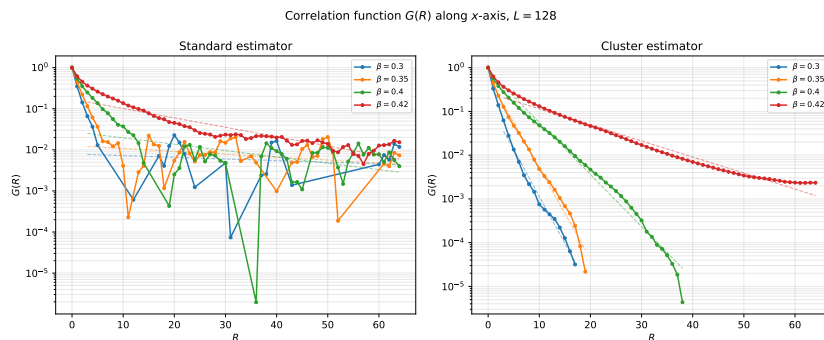


Figure 2: Correlation function $G(R)$ along the x -axis for $L = 128$.

The correlation function decreases exponentially with distance for all temperatures considered,

$$G(R) \propto e^{-R/\xi}.$$

The cluster estimator shows considerably smaller fluctuations than the standard estimator, particularly at large distances where $G(R)$ is small. This leads to a more accurate determination of the correlation length.

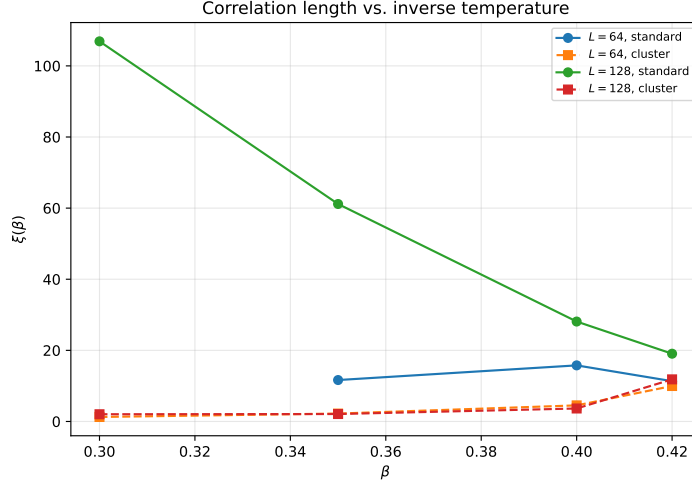


Figure 3: Correlation length as a function of inverse temperature.

As expected, the correlation length increases strongly with β and becomes largest close to the critical point. The results obtained for $L = 64$ and $L = 128$ are consistent and indicate only small finite-size effects in the investigated temperature range.

2 Statistical error of histograms

For a 16×16 lattice at $\beta_c \approx 0.440686$, the Metropolis algorithm was used with 5000 sweeps for thermalisation and $2^{16} = 65536$ sweeps for measurements.

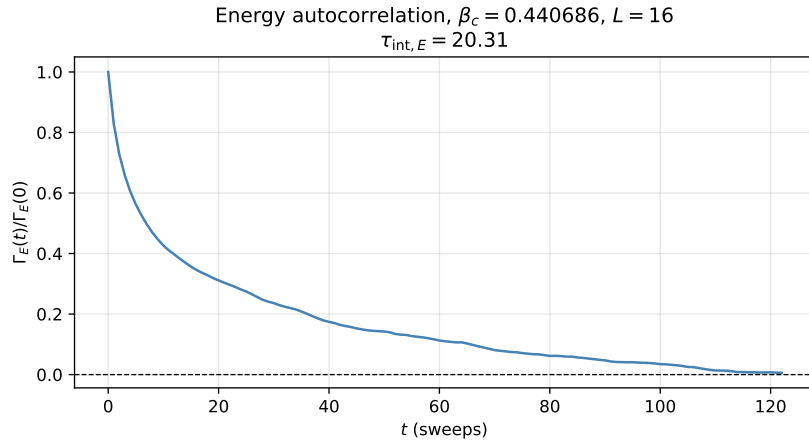


Figure 4: Energy autocorrelation function.

The integrated autocorrelation time obtained from the energy autocorrelation function is

$$\tau_{\text{int},E} = 20.31.$$

The measured energy histogram agrees with the expected distribution at criticality. The histogram errors were obtained from 16 blocks and compared with the Poisson estimate

$$\sigma_{H(E)} \simeq \sqrt{H(E)}.$$

The measured histogram errors are significantly larger than the Poisson prediction due to autocorrelations in the Markov chain. From the measured autocorrelation time we expect an enhancement factor

$$\sqrt{2\tau_{\text{int},E}} = \sqrt{2 \cdot 20.31} = 6.37,$$

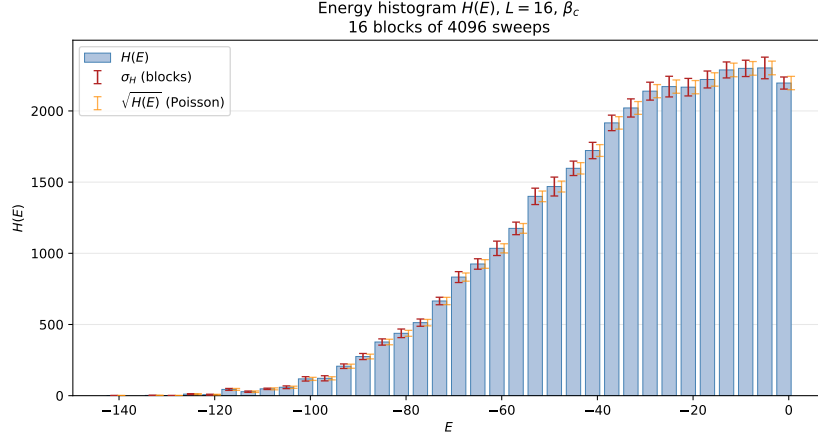


Figure 5: Energy histogram $H(E)$.

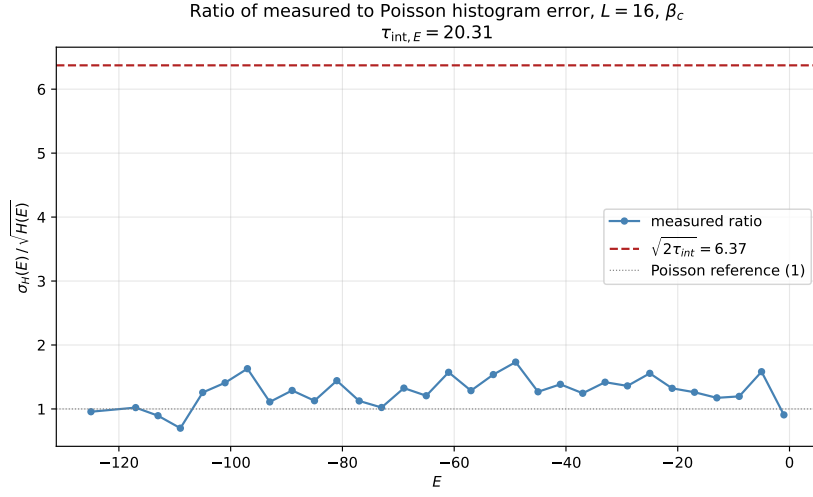


Figure 6: Ratio of measured histogram error to the Poisson estimate.

which is consistent with the observed ratio. Therefore the histogram errors are larger than $\sqrt{H(E)}$ by approximately the expected factor arising from autocorrelated measurements.

HW8 - Computer Simulation II

Schewtschik, Guilherme (3748052)

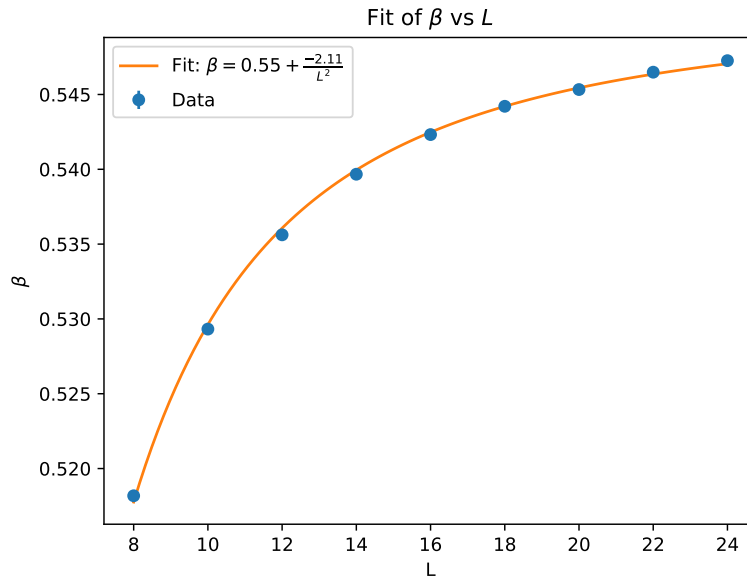
June 9, 2026

1

Given that beta and L are related by the equation:

$$\beta = a + \frac{b}{L^2} \quad (1)$$

It is best to work with the variable $\tilde{L} = 1/L^2$, and then convert it back to L . Fitting a and b for $L_{min} = 8$ we get the following graph:



We can see that the fit is at a first glance good but deviates in some points, lets calculate the relative error of the individual points and $\chi^2/(N - 2)$:

$$\frac{\chi^2}{N - 2} \approx 10.44$$

β	β_{fit}	Relative Error
0.51818	0.51776	4.22
0.52932	0.52962	-3.03
0.53562	0.53607	-4.07
0.53967	0.53995	-2.84
0.54232	0.54248	-1.95
0.54421	0.54421	0.05
0.54533	0.54544	-1.41
0.54649	0.54636	2.64
0.54726	0.54705	2.94

Table 1: Fit results for all lattice sizes $L = 8, \dots, 24$.

Given that the ansatz expects $\chi^2/(N - 2) \approx 1$, the fit quality is poor for $L_{min} = 8$, now ranging over L_{min} values we can find the optimal range of L values to fit the data:

$L_{\min}$	N	a	b	χ^2/dof
8	9	0.550716	-2.109338	10.443
10	8	0.550936	-2.181172	2.859
12	7	0.551079	-2.234493	1.143
14	6	0.551122	-2.251716	1.268
16	5	0.551162	-2.269470	1.584
18	4	0.551242	-2.307417	2.210

Table 2: Weighted fits for different lower cutoffs $L_{\min}$.

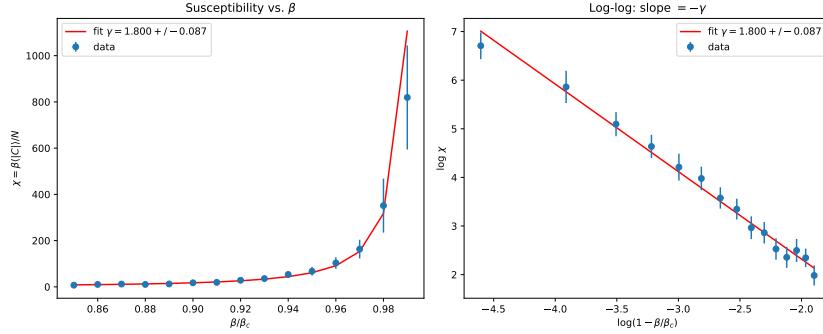
So for $L_{\min} = 12$ we get the best fit, with $\chi^2/(N - 2) \approx 1.143$, and the fit parameters are $a \approx 0.55$ and $b \approx -2.23$. This means that the critical beta value in the thermodynamic limit is approximately $\beta_c \approx 0.551079$.

2

Rewrite the equation in a logarithmic form:

$$\ln(\chi(\beta)) = -\gamma \ln(a) - \gamma \ln(1 - \beta/\beta_c)$$

Fitting the curve in log space for $\beta \in [0.99, 0.85]\beta_c$ and taking into account the error $e_{\ln(\chi)} = \sigma/\chi$, where sigma is the standard deviation from cluster size, we get the following plots.



The values for $\gamma = 1.80 \pm 0.087$ is well in the theoretical expectation of $\gamma = 1.75$ for the 2D Ising model.

HW9 - Computer Simulation II

Schewtschik, Guilherme (3748052)

June 15, 2026

1

Using the histogram reweighting method we get

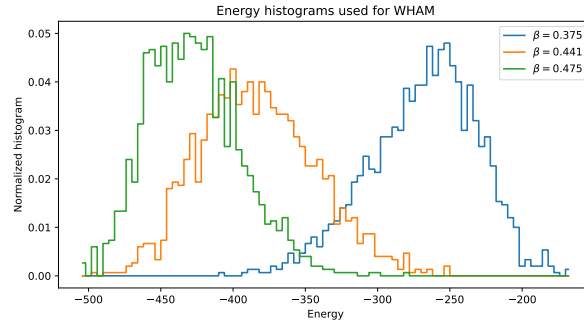


Figure 1: Energy histograms.

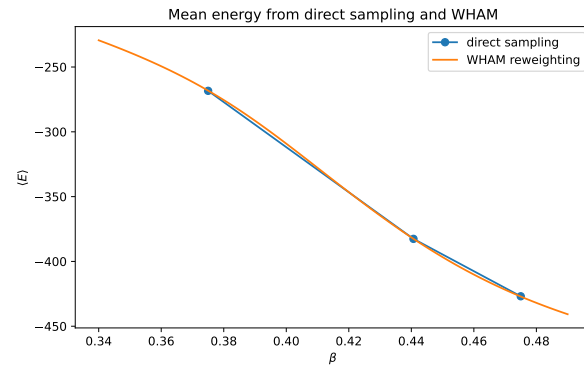


Figure 2: Mean energy from direct sampling and from WHAM.

The WHAM estimate agrees well with the direct measurements near the simulated values and provides a smooth interpolation of the energy curve between the three simulation points.

2

Using the improve estimator for the susceptibility we get:

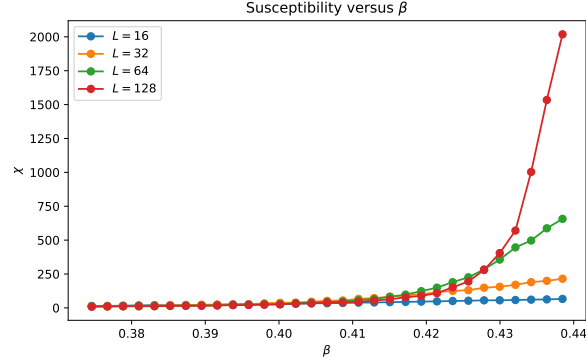


Figure 3: Susceptibility as a function of β for several lattice sizes.

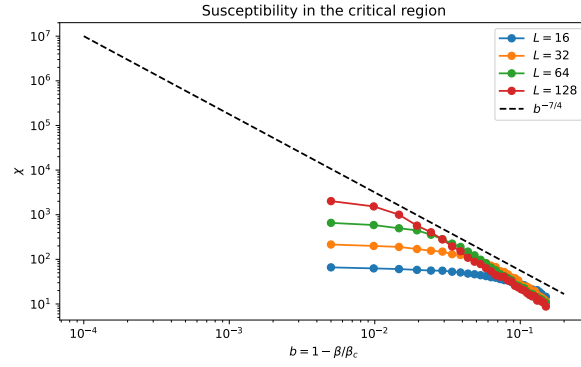


Figure 4: Log-log plot of χ versus $b = 1 - \beta/\beta_c$ with the expected power law $b^{-7/4}$.

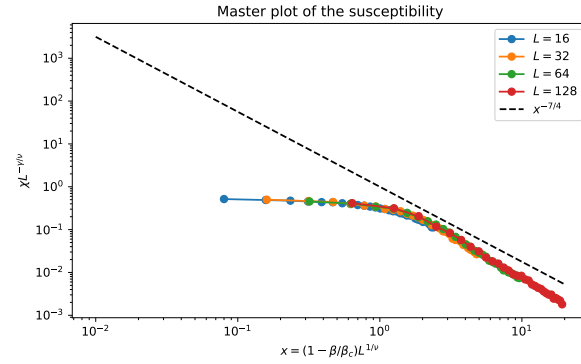


Figure 5: Master plot of $\chi L^{-\gamma/\nu}$ versus $x = (1 - \beta/\beta_c)L^{1/\nu}$, using $\nu = 1$ and $\gamma = 7/4$.

The susceptibility grows strongly near the critical point $\beta = 0.44$ for $L > 16$, scaling with lattice size. In the second plot we can see that the susceptibility approaches the expected near β_c but still shows big dependence on lattice size, while the master plot shows a lattice size independent result.

19.

Given $U_4 = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2}$

$$\frac{dU_4}{d\beta} = -\frac{1}{3} \frac{d}{d\beta} \left(\frac{\langle m^4 \rangle}{\langle m^2 \rangle^2} \right) = -\frac{1}{3} \left[\frac{d}{d\beta} (\langle m^4 \rangle) \frac{1}{\langle m^2 \rangle^2} - \frac{2\langle m^4 \rangle}{\langle m^2 \rangle^3} \frac{d}{d\beta} (\langle m^2 \rangle) \right]$$

$$\frac{d\langle m^4 \rangle}{d\beta} = \frac{d}{d\beta} \left(\frac{\sum m^4 e^{-\beta H}}{\sum e^{-\beta H}} \right) = \frac{\sum -m^4 H e^{-\beta H}}{\sum e^{-\beta H}} - \frac{\sum m^4 e^{-\beta H} \sum -H e^{-\beta H}}{(\sum e^{-\beta H})^2}$$

$= -\langle m^4 H \rangle + \langle m^4 \rangle \langle H \rangle$

$$\Rightarrow \frac{d\langle m^4 \rangle}{d\beta} = -\langle m^4 H \rangle + \langle m^4 \rangle \langle H \rangle$$

$$\frac{2\langle m^4 \rangle}{\langle m^2 \rangle^3} \frac{d\langle m^2 \rangle}{d\beta} = \frac{2\langle m^4 \rangle}{\langle m^2 \rangle^2} \frac{d \ln(\langle m^2 \rangle)}{d\beta}$$

$$\Rightarrow \frac{2\langle m^4 \rangle}{\langle m^2 \rangle^3} \frac{d\langle m^2 \rangle}{d\beta} = -\frac{2\langle m^4 \rangle \langle m^2 H \rangle}{\langle m^2 \rangle^3} + \frac{2\langle m^4 \rangle \langle H \rangle \langle m^2 \rangle}{\langle m^2 \rangle^3}$$

$$\Rightarrow \frac{dU_4}{d\beta} = -\frac{1}{3} \left[\frac{-\langle m^4 H \rangle + \langle m^4 \rangle \langle H \rangle}{\langle m^2 \rangle^2} - \frac{2\langle H \rangle \langle m^4 \rangle}{\langle m^2 \rangle^2} + \frac{2\langle m^4 \rangle \langle m^2 H \rangle}{\langle m^2 \rangle^3} \right]$$

$$\frac{dU_4}{d\beta} = -\frac{1}{3} \left[\frac{1}{\langle m^2 \rangle^2} \left[\frac{2\langle m^4 \rangle \langle m^2 H \rangle}{\langle m^2 \rangle} - \langle m^4 H \rangle - \langle H \rangle \langle m^4 \rangle \right] \right]$$

$$\Rightarrow \frac{dU_4}{d\beta} = -\frac{1}{3} \left[\frac{\langle m^4 \rangle}{\langle m^2 \rangle^2} \left[\frac{2\langle m^2 H \rangle}{\langle m^2 \rangle} - \frac{\langle m^4 H \rangle}{\langle m^4 \rangle} - \langle H \rangle \right] \right]$$

$$\frac{dU_4}{d\beta} = \frac{1}{3} \frac{\langle m^4 \rangle}{\langle m^2 \rangle^2} \left(\langle H \rangle - 2 \frac{\langle m^2 H \rangle}{\langle m^2 \rangle} + \frac{\langle m^4 H \rangle}{\langle m^4 \rangle} \right)$$

$$\frac{1}{3} \frac{\langle m^2 \rangle}{\langle m^2 \rangle^2} = 1 - u_1$$

$$\Rightarrow \frac{d u_1}{d \beta} = (1 - u_1) \left(\langle H \rangle - 2 \frac{\langle m^2 H \rangle}{\langle m^2 \rangle} + \frac{\langle m^4 H \rangle}{\langle m^4 \rangle} \right)$$

For a Hamiltonian $H = eV$ where e is the energy densities and V the volume" ($V = L^2$)

$$\langle H \rangle = V \langle e \rangle$$

$$\langle H m^2 \rangle = V \langle A e \rangle$$

$$\Rightarrow \frac{d u_1}{d \beta} = V(1 - u_1) \left(\langle e \rangle - 2 \frac{\langle m^2 e \rangle}{\langle m^2 \rangle} + \frac{\langle m^4 e \rangle}{\langle m^4 \rangle} \right)$$

For $u_2 = 1 - \frac{\langle m^2 \rangle}{3 \langle m^4 \rangle}$ the result follows the same form.

$$\frac{d u_2}{d \beta} = V(1 - u_2) \left(\langle e \rangle - \frac{2 \langle m^2 e \rangle}{\langle m^2 \rangle} + \frac{\langle m^4 e \rangle}{\langle m^4 \rangle} \right)$$

b)

$$\frac{d \ln(\langle m^2 \rangle)}{d \beta} = \frac{d \langle m^2 \rangle}{d \beta} \frac{1}{\langle m^2 \rangle} = \frac{1}{\langle m^2 \rangle} \left(\langle m^2 \rangle \langle H \rangle - \langle m^2 H \rangle \right)$$

$$\frac{d \ln(\langle m^2 \rangle)}{d \beta} = \langle H \rangle - \frac{\langle m^2 H \rangle}{\langle m^2 \rangle} = V \left(\langle e \rangle - \frac{\langle m^2 e \rangle}{\langle m^2 \rangle} \right)$$

Similarly for $\langle m^4 \rangle$

$$\frac{d \ln \langle m^4 \rangle}{d \beta} = V \left(\langle e \rangle - \frac{\langle m^4 e \rangle}{\langle m^4 \rangle} \right)$$

HW10 - Computer Simulation II

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June 22, 2026

Problem 20

At a second-order transition, the different observables at $\beta = \beta_c$ scale as:

$$\chi(L) \sim L^{\gamma/\nu}, \quad C(L) \sim L^{\alpha/\nu}, \quad \left| \frac{dU_4}{d\beta} \right|_{\beta_c} \sim L^{1/\nu}, \quad (1)$$

with exact exponents $\gamma/\nu = 7/4$, $\alpha/\nu = 0$, and $1/\nu = 1$ for the Ising universal class. The Binder cumulants U_4 and U_2 converge to constants in the thermodynamic limit at $L \rightarrow \infty$.

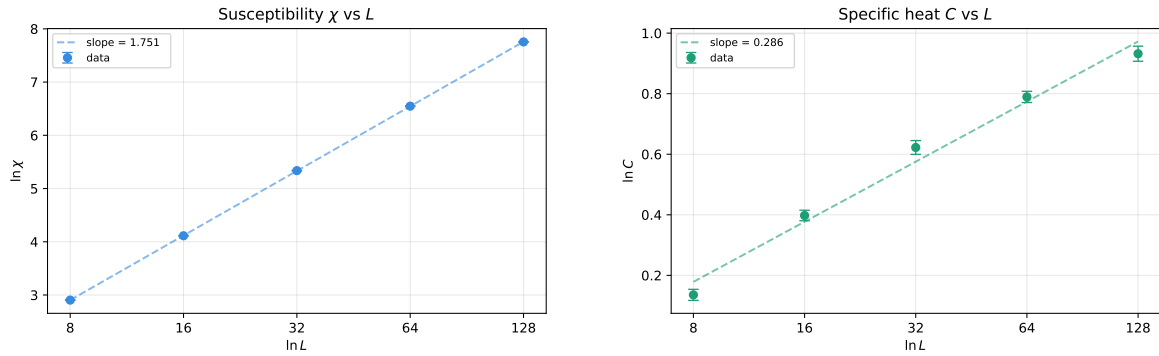


Figure 1: Left: susceptibility χ vs L in log-log scale. Right: specific heat C vs L .

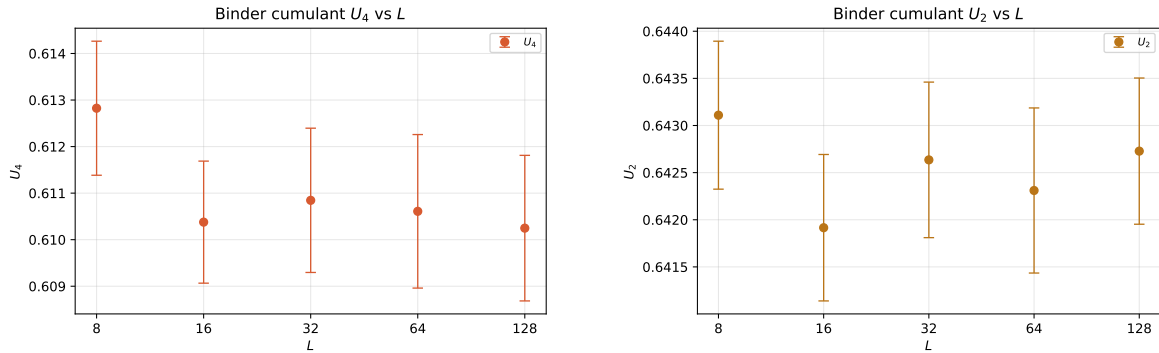


Figure 2: Left: Binder cumulants U_4 . Right: U_2 (right) vs L .

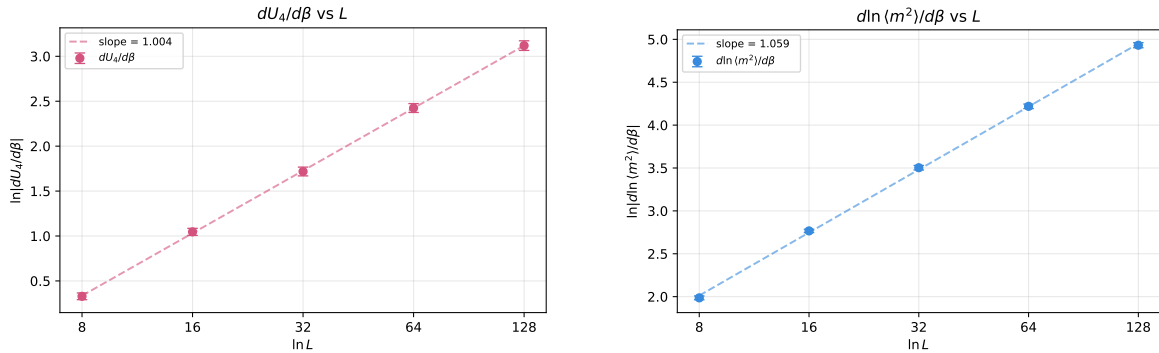


Figure 3: Left: Derivatives $dU_4/d\beta$. Right: $d \ln(m^2)/d\beta$ vs L in log-log scale.

Table 1: Measured observables at β_c for each lattice size L . Errors are standard deviation estimates.

L	C	χ	U_4	U_2	$dU_4/d\beta$	$d\ln\langle m^2 \rangle/d\beta$
8	1.13	18.2	0.612 ± 0.002	0.644 ± 0.002	1.40 ± 0.06	7.29 ± 0.41
16	1.46	61.8	0.613 ± 0.002	0.646 ± 0.002	2.66 ± 0.14	14.08 ± 1.02
32	1.87	204.1	0.608 ± 0.002	0.641 ± 0.002	5.91 ± 0.31	28.30 ± 2.80
64	2.16	694.2	0.609 ± 0.002	0.642 ± 0.002	11.74 ± 0.57	57.99 ± 4.12
128	2.65	2320	0.608 ± 0.002	0.641 ± 0.002	23.72 ± 1.18	118.1 ± 9.9

Table 2: Fitted log-log slopes versus exact FSS predictions.

Quantity	Fitted slope	Exact	Deviation
$\chi \sim L^{\gamma/\nu}$	1.749	1.750	0.06%
$C \sim L^{\alpha/\nu}$ (log divergence)	0.201	0 (log)	-
$dU_4/d\beta \sim L^{1/\nu}$	1.032	1.000	3.2%
$d\ln\langle m^2 \rangle/d\beta \sim L^{1/\nu}$	1.011	1.000	1.1%

The susceptibility exponent $\gamma/\nu = 1.749 \pm 0.003$ agrees with the exact value $7/4$. The specific heat grows slowly with L , consistent with the known logarithmic divergence ($\alpha = 0$) rather than a power law. Both derivative quantities scale as $L^{1/\nu}$ with $1/\nu \approx 1.02$, confirming $\nu = 1$ and validating the results from Problem 19.

The Binder cumulant U_4 hovers around 0.61 across all lattice sizes, consistent with the known universal value $U_4^* \approx 0.611$. The small remaining L dependence is a finite size correction of order L^{-2} .

HW9 - Computer Simulation II

Schewtschik, Guilherme (3748052)

June 29, 2026

1

Using the following function to reweigh the observables:

```
def reweight_moments(energies, magnetizations, L,
                    beta0, beta_values):
    V = L * L

    m = magnetizations / V

    absm = np.abs(m)

    m2 = m ** 2
    m4 = m ** 4

    results = []

    for beta in beta_values:

        delta = beta - beta0

        # logarithmic weights improve numerical stability
        logw = -delta * energies
        logw -= np.max(logw)

        w = np.exp(logw)

        Z = np.sum(w)

        mean_absm = np.sum(w * absm) / Z
        mean_m2 = np.sum(w * m2) / Z
        mean_m4 = np.sum(w * m4) / Z

        results.append(
            {
                "beta": beta,
                "absm": mean_absm,
                "m2": mean_m2,
                "m4": mean_m4,
            }
        )

    return pd.DataFrame(results)
```


We obtain the following graph:

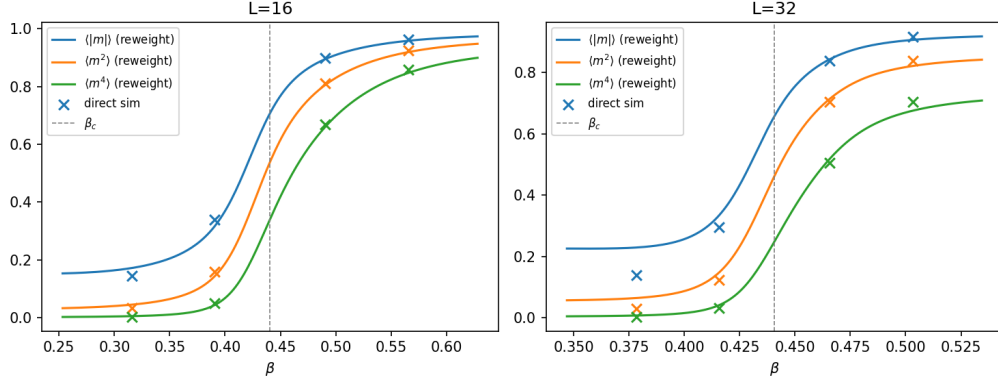


Figure 1: Magnetizations.

We can see the reweighted observables stay close to the actual simulation and display the expected behaviour becoming small below β_c and approaching 1 above the critical temperature, we also see that the higher powers of the magnetization stay closer to zero in the unorderd phase and need a higher temperature to truly magnetize. The size scaling shows that with bigger lattice sizes this effect is even more exaggerated, where the different powers of the magnetization stay separate for longer in the $L = 32$ case.

2

Simulating the potts model for $L = 16, 32, 64, 128$, and using the binning method to estimate the auto correlation time

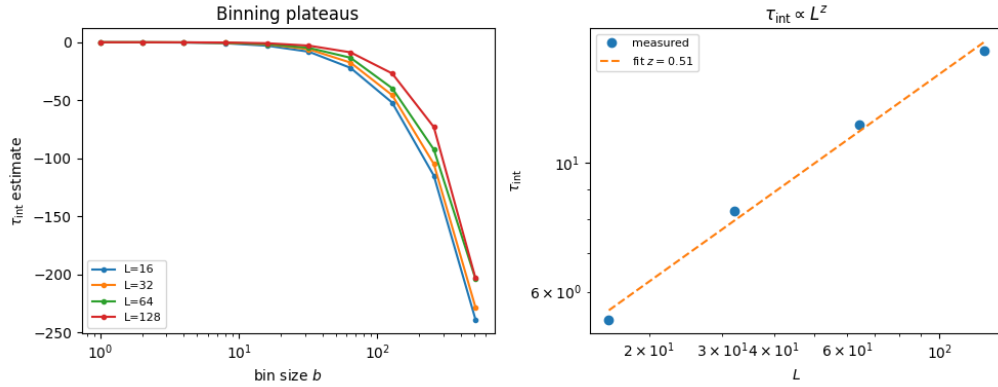


Figure 2: Autocorrelation time.

The binning plateaus confirm that τ_{int} is small for all system sizes. A log-log fit of τ_{int} and L yields the value for $z \approx 0.51$. The fitted z value is noticeably larger than the value observed for the Wolff algorithm on the 2D Ising model of $z \approx 0.25$, which is expected.